

Every Formula, Chapters 1 · 2 · 3 · 4 · 7

All formulae, constants, standard values, shapes and resonance structures from the five chapters, in one place. Items marked † are standard exam formulae not stated in the NCERT text itself.

Universal constants. $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$ · $h = 6.626 \times 10^{-34} \text{ J s}$ · $c = 3.0 \times 10^8 \text{ m s}^{-1}$ · $m_e = 9.11 \times 10^{-31} \text{ kg}$ · $e = 1.602 \times 10^{-19} \text{ C}$ · $R_H = 109,677 \text{ cm}^{-1}$ · $1 \text{ u} = 1.66 \times 10^{-24} \text{ g}$ · $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$ · $1 \text{ bar} = 10^5 \text{ Pa}$ · $1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$

Chapter 1 · Some Basic Concepts of Chemistry

1 · The mole — the three bridges

MASS ↔ MOLE

$$n = \frac{m}{M}$$

m in g, M in g mol⁻¹

PARTICLES ↔ MOLE

$$n = \frac{N}{N_A}$$

$N_A = 6.022 \times 10^{23}$

GAS VOLUME ↔ MOLE (STP)

$$n = \frac{V}{22.7 \text{ L}}$$

22.7 L at 273.15 K & 1 bar; 22.4 L at 1 atm

SOLUTION ↔ MOLE

$$n = M \times V_{(L)}$$

M = molarity

2 · Masses and formulae

AVERAGE ATOMIC MASS

$$\bar{A} = \sum (\text{fractional abundance} \times \text{isotopic mass})$$

MASS % OF AN ELEMENT

$$\% E = \frac{\text{mass of E in 1 mol}}{\text{molar mass}} \times 100$$

MOLECULAR FROM EMPIRICAL

$$\text{MF} = (\text{EF})_n, \quad n = \frac{M}{\text{EFM}}$$

LIMITING REAGENT TEST

smallest $\frac{\text{moles}}{\text{coefficient}}$
never compare raw masses

% yield = $\frac{\text{actual yield}}{\text{theoretical yield}} \times 100$, theoretical yield being what the limiting reagent predicts.

3 • Concentration of solutions

Term	Formula	Unit	Depends on T?
Mass percentage	$\frac{w_B}{w_A + w_B} \times 100$	%	No
Mole fraction	$x_B = \frac{n_B}{n_A + n_B}$; $x_A + x_B = 1$	—	No
Molarity	$M = \frac{n_B}{V_{\text{solution}}(\text{L})}$	mol L ⁻¹	Yes
Molality	$m = \frac{n_B}{w_A(\text{kg})}$	mol kg ⁻¹	No
ppm	$\frac{\text{mass of solute}}{\text{mass of solution}} \times 10^6$	ppm	No
Normality †	$N = M \times n\text{-factor}$	eq L ⁻¹	Yes

DILUTION

$$M_1 V_1 = M_2 V_2$$

MIXING TWO SOLUTIONS †

$$M_f = \frac{M_1 V_1 + M_2 V_2}{V_1 + V_2}$$

MOLARITY FROM MASS % AND DENSITY

$$M = \frac{10 \times (\text{mass \%}) \times d}{M_B}$$

d in g mL⁻¹

MOLARITY → MOLALITY †

$$m = \frac{1000 M}{1000 d - M M_B}$$

4 • Units, significant figures, temperature

$$T(\text{K}) = t(^{\circ}\text{C}) + 273.15 \quad \cdot \quad t(^{\circ}\text{F}) = \frac{9}{5}t(^{\circ}\text{C}) + 32 \quad \cdot \quad 1 \text{ L} = 1 \text{ dm}^3 = 10^3 \text{ cm}^3 = 10^{-3} \text{ m}^3 \quad \cdot \quad \text{density}$$

SI unit kg m⁻³

Significant figures: addition/subtraction keeps the fewest **decimal places**;
multiplication/division keeps the fewest **significant figures**.

Chapter 2 • Structure of Atom

1 • Sub-atomic particles

Quantity	Value	Quantity	Value
e/m_e (Thomson)	$1.758820 \times 10^{11} \text{ C kg}^{-1}$	Mass of proton	$1.67262 \times 10^{-27} \text{ kg}$
e (Millikan)	$1.602 \times 10^{-19} \text{ C}$	Mass of neutron	$1.67493 \times 10^{-27} \text{ kg}$
Mass of electron	$9.1094 \times 10^{-31} \text{ kg}$	Neutrons	$A - Z$

2 • Radiation and the quantum

WAVE RELATION

$$c = \nu \lambda \quad \cdot \quad \bar{\nu} = \frac{1}{\lambda}$$

PLANCK / PHOTON ENERGY

$$E = h\nu = \frac{hc}{\lambda} \quad \cdot \quad E = nh\nu$$

PHOTOELECTRIC EQUATION

$$h\nu = h\nu_0 + \frac{1}{2}m_e v^2$$

$$KE_{\max} = h(\nu - \nu_0) ; \text{work function } \phi = h\nu_0$$

STOPPING POTENTIAL †

$$eV_0 = h(\nu - \nu_0)$$

EV SHORTCUT †

$$E(\text{eV}) = \frac{1240}{\lambda(\text{nm})}$$

RYDBERG FORMULA

$$\bar{\nu} = 109677 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ cm}^{-1}$$

Series	n_1	Region	Series	n_1	Region
Lyman	1	UV	Brackett	4	IR
Balmer	2	Visible	Pfund	5	Far IR
Paschen	3	IR	Lines to ground state = $\frac{n(n-1)}{2}$ †		

3 • Bohr's model — results and the derivation chain**RADIUS OF NTH ORBIT**

$$r_n = 52.9 \frac{n^2}{Z} \text{ pm}$$

$$r \propto n^2, r \propto 1/Z$$

ENERGY OF NTH ORBIT

$$E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$$

$$= -2.18 \times 10^{-18} \frac{Z^2}{n^2} \text{ J}$$

QUANTISED ANGULAR MOMENTUM

$$mvr = \frac{nh}{2\pi}$$

VELOCITY IN NTH ORBIT †

$$v_n = 2.188 \times 10^6 \frac{Z}{n} \text{ m s}^{-1}$$

FREQUENCY RULE

$$\Delta E = E_2 - E_1 = h\nu$$

ENERGY PARTITION †

$$PE = 2E \cdot KE = -E$$

DERIVATION CHAIN † Bohr radius and energy, in the order you write them

$$1. \text{Coulomb attraction} = \text{centripetal force: } \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r^2} = \frac{mv^2}{r} \Rightarrow mv^2 = \frac{Ze^2}{4\pi\epsilon_0 r} \dots(i)$$

$$2. \text{Quantisation: } mvr = \frac{nh}{2\pi} \Rightarrow v = \frac{nh}{2\pi mr} \dots(ii)$$

$$3. (ii) \text{ into } (i): \frac{n^2 h^2}{4\pi^2 m r^2} = \frac{Ze^2}{4\pi\epsilon_0 r} \Rightarrow \boxed{r = \frac{n^2 h^2 \epsilon_0}{\pi m Z e^2}}$$

$$4. KE = \frac{1}{2}mv^2 = \frac{Ze^2}{8\pi\epsilon_0 r} ; PE = -\frac{Ze^2}{4\pi\epsilon_0 r}$$

$$5. E = KE + PE = -\frac{Ze^2}{8\pi\epsilon_0 r} \Rightarrow \boxed{E_n = -\frac{mZ^2 e^4}{8\epsilon_0^2 h^2 n^2}}$$

$$6. \text{Transition: } \Delta E = 2.18 \times 10^{-18} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ J} \Rightarrow \bar{\nu} = \frac{\Delta E}{hc} = 109677 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ cm}^{-1}$$

4 · de Broglie, Heisenberg, quantum numbers

DE BROGLIE WAVELENGTH

$$\lambda = \frac{h}{mv} = \frac{h}{p}$$

ACCELERATED ELECTRON †

$$\lambda = \frac{12.27}{\sqrt{V}} \text{ Å}$$

UNCERTAINTY PRINCIPLE

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

$$\Delta x \cdot \Delta v \geq \frac{h}{4\pi m}$$

ORBITAL ANGULAR MOMENTUM

$$\sqrt{l(l+1)} \frac{h}{2\pi}$$

zero for every s orbital

Quantum number	Allowed values	Counting rule	Value
Principal n	1, 2, 3 ...	Electrons in a shell	$2n^2$
Azimuthal l	0 to $(n-1)$	Orbitals in a subshell	$2l+1$
Magnetic m_l	$-l \dots 0 \dots +l$	Electrons in a subshell	$2(2l+1)$
Spin m_s	$+\frac{1}{2}$ or $-\frac{1}{2}$	Filling order	lower $(n+l)$ first; tie $\rightarrow$ lower n

Nodes. Angular (planar) = l · Radial (spherical) = $n - l - 1$ · **Total** = $n - 1$ — the first two must add to the third.

Aufbau order: 1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f 5d 6p 7s. **Exceptions:** Cr = [Ar]3d⁵4s¹, Cu = [Ar]3d¹⁰4s¹. **On ionisation 4s empties before 3d** (Fe²⁺ = [Ar]3d⁶).

Chapter 3 · Classification of Elements and Periodicity

Modern Periodic Law. The physical and chemical properties of the elements are periodic functions of their **atomic numbers**. **18 groups, 7 periods.** Period number = highest n in the configuration.

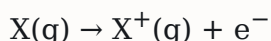
Elements per period: 2, 8, 8, 18, 18, 32, 32 · filled as 1s | 2s2p | 3s3p | 4s3d4p | 5s4d5p | 6s4f5d6p | 7s5f6d7p

Block	Groups	General configuration	Key features
s	1, 2	ns^{1-2}	Reactive metals, low IE, +1/+2, strong reducing agents
p	13-18	$ns^2 np^{1-6}$	Metals, non-metals and metalloids; group 18 noble gases
d	3-12	$(n-1)d^{1-10} ns^{0-2}$	Coloured ions, variable oxidation states, catalytic, paramagnetic. Zn/Cd/Hg excluded
f	separate	$(n-2)f^{1-14} (n-1)d^{0-1} ns^2$	Lanthanoids 58-71, actinoids 90-103 (radioactive); mainly +3

Trend summary — the whole chapter in one table

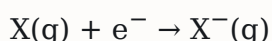
Property	Across a period →	Down a group ↓	Reason
Atomic radius	Decreases	Increases	Nuclear charge rises with no new shell / new shell added
Ionization enthalpy	Increases	Decreases	Size and shielding
Electron gain enthalpy	More negative	Less negative	Size
Electronegativity	Increases	Decreases	Size and nuclear charge
Metallic character	Decreases	Increases	Ease of losing electrons
Nature of oxide	Basic → amphoteric → acidic	More basic	—

IONIZATION ENTHALPY



always **positive**; $\Delta_i H_1 < \Delta_i H_2 < \Delta_i H_3$

ELECTRON GAIN ENTHALPY



negative if energy released; **2nd value always positive**

THE ANOMALIES — MEMORISE THESE FOUR LINES

1. **Be > B** and **Mg > Al** (ionization): the outgoing electron is from a higher-energy, better-shielded **p** orbital, versus breaking a filled **s²**.
2. **N > O** and **P > S** (ionization): half-filled **p³** is extra stable and symmetrical; the paired p electron in O suffers repulsion.
3. **Cl > F** and **S > O** (electron gain, more negative): F and O are so small that the compact 2p subshell causes strong **inter-electronic repulsion**.
4. **Isoelectronic radius** falls as nuclear charge rises: $O^{2-}(140) > F^-(136) > Na^+(95) > Mg^{2+}(65) > Al^{3+}(50)$ pm.

Period 2	Li	Be	B	C	N	O	F	Ne
Atomic radius / pm	152	111	88	77	74	66	64	—
$\Delta_i H_1$ / kJ mol ⁻¹	520	899	801	1086	1402	1314	1681	2080

Electron gain enthalpies (kJ mol^{-1}): F -328 , Cl -349 , Br -325 , I -295 , O -141 , S -200 . **Group 1 radii** (pm): Li 152, Na 186, K 231, Rb 244, Cs 262. **Electronegativity**: Pauling scale, F = 4.0 highest.

IUPAC roots ($Z > 100$): 0 nil · 1 un · 2 bi · 3 tri · 4 quad · 5 pent · 6 hex · 7 sept · 8 oct · 9 enn, then **-ium**. 104 = unnilquadium (Unq) · 120 = unbinilium (Ubn).

Second-period anomaly: small size, high charge/radius ratio and electronegativity, **no d orbitals** $\Rightarrow$ max covalency 4. **Diagonal pairs**: Li-Mg, Be-Al, B-Si.

Period 3 oxides: Na_2O , MgO (basic) · **Al_2O_3 (amphoteric)** · SiO_2 , P_4O_{10} , SO_3 , Cl_2O_7 (acidic). Neutral: CO, NO, N_2O .

Chapter 4 · Chemical Bonding and Molecular Structure

FORMAL CHARGE

$$\text{FC} = V - L - \frac{1}{2}B$$

valence – lone-pair electrons $-\frac{1}{2}(\text{bonding electrons})$

HYBRIDISATION / STERIC NUMBER [†]

$$H = \frac{1}{2} [V + M - C + A]$$

M = monovalent atoms; C cation charge; A anion charge

BOND ORDER (MOT)

$$\text{B.O.} = \frac{N_b - N_a}{2}$$

B.O. $\uparrow \Rightarrow$ bond enthalpy $\uparrow$, bond length $\downarrow$

DIPOLE MOMENT

$$\mu = q \times d$$

1 D = 3.336×10^{-30} C m; vector sum

Shapes — VSEPR geometry vs molecular shape



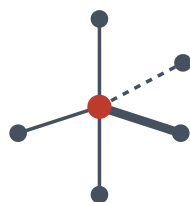
Linear · 180°
 BeCl_2 , CO_2



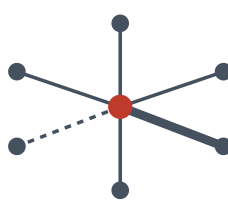
Trigonal planar · 120°
 BF_3 , BCl_3



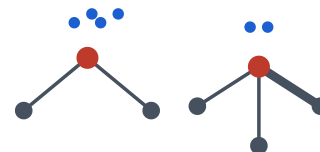
Tetrahedral · 109.5°
 CH_4 , NH_4^+



Trigonal bipyramidal
 PCl_5 · 90° and 120°



Octahedral · 90°
 SF_6



Bent & pyramidal
 H_2O 104.5° · NH_3 107°

The five basic geometries plus the two commonest lone-pair shapes. Blue dots are lone pairs. Geometry counts all electron pairs; shape is what remains when only atoms are drawn.

Type	BP	LP	Hybrid	Geometry	Shape	Example (angle)
AB ₂	2	0	sp	Linear	Linear	BeCl ₂ , CO ₂ (180°)
AB ₃	3	0	sp ²	Trigonal planar	Trigonal planar	BF ₃ (120°)
AB ₂ E	2	1	sp ²	Trigonal planar	Bent	SO ₂ (119.5°)
AB ₄	4	0	sp ³	Tetrahedral	Tetrahedral	CH ₄ (109.5°)
AB ₃ E	3	1	sp ³	Tetrahedral	Trigonal pyramidal	NH ₃ (107°)
AB ₂ E ₂	2	2	sp ³	Tetrahedral	Bent	H ₂ O (104.5°)
AB ₅	5	0	sp ³ d	Trigonal bipyramidal	Trigonal bipyramidal	PCl ₅ (90°, 120°)
AB ₄ E	4	1	sp ³ d	Trigonal bipyramidal	See-saw	SF ₄
AB ₃ E ₂	3	2	sp ³ d	Trigonal bipyramidal	T-shaped	ClF ₃
AB ₂ E ₃	2	3	sp ³ d	Trigonal bipyramidal	Linear	XeF ₂ , I ₃ ⁻
AB ₆	6	0	sp ³ d ²	Octahedral	Octahedral	SF ₆ (90°)
AB ₅ E	5	1	sp ³ d ²	Octahedral	Square pyramidal	BrF ₅
AB ₄ E ₂	4	2	sp ³ d ²	Octahedral	Square planar	XeF ₄

Repulsion order lp-lp > lp-bp > bp-bp. Lone pairs take **equatorial** positions in a trigonal bipyramid and go **trans** in an octahedron.

Bond angles: CH₄ 109.5° > NH₃ 107° > H₂O 104.5° · NH₃ 107° > NF₃ 102° · NH₃ 107° > PH₃ 93.6°

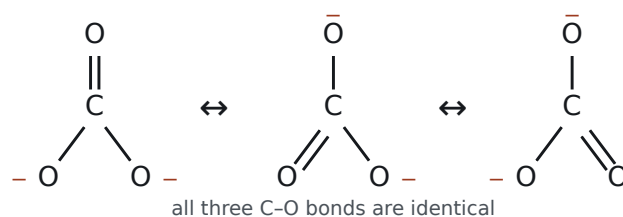
Dipole moments (D): BeF₂ 0 · CO₂ 0 · BF₃ 0 · CCl₄ 0 · **H₂O 1.85** · **NH₃ 1.47** · **NF₃ 0.24** · HF 1.78

Bond length / enthalpy: C-C 154 pm > C=C 133 > C≡C 120 · N≡N 946 > O=O 498 > H-H 435.8 kJ mol⁻¹ · lattice enthalpy NaCl 788 kJ mol⁻¹

Resonance structures



Ozone, O₃



Carbonate ion, CO₃²⁻

Rules. Canonical forms differ only in the **positions of electrons**, never of nuclei. They have **no independent existence** — only the hybrid is real, and it is **lower in energy** than any single form. All bonds in the hybrid are **equivalent**: O₃ both bonds 128 pm (between single 148 and double 121); CO₃²⁻ and NO₃⁻ all bonds identical — the nitrate ion has exactly the same three-form pattern as the carbonate ion drawn above.

Molecular orbital theory

Energy order for B₂, C₂, N₂ (≤ 14 electrons):



For O₂, F₂, Ne₂ (> 14 electrons): the last two swap — $\sigma 2p_z < \pi 2p_x = \pi 2p_y$

Species	H ₂	He ₂	Li ₂	Be ₂	B ₂	C ₂	N ₂	O ₂	F ₂	Ne ₂
Bond order	1	0	1	0	1	2	3	2	1	0
Magnetic	dia	—	dia	—	para	dia	dia	para	dia	—

Oxygen series	O ₂ ⁺	O ₂	O ₂ [−]	O ₂ ^{2−}
Valence electrons	11	12	13	14
Bond order	2.5	2	1.5	1
Unpaired electrons	1	2	1	0

Bond length O₂⁺ < O₂ < O₂[−] < O₂^{2−}; stability runs the other way. Added electrons enter **antibonding** orbitals.

Fajans' rules — covalent character rises with: small cation · large anion · high charge · pseudo-noble-gas (18-electron) cation.

Octet failures: incomplete (BCl₃, BeH₂) · expanded (PF₅, SF₆) · odd-electron (NO, NO₂) · noble gas compounds (XeF₂).

Hydrogen bond 10–40 kJ mol^{−1}, needs H bonded to **F, O or N**. Intermolecular **raises** boiling point; intramolecular (*o*-nitrophenol) **lowers** it. Strength HF > H₂O > NH₃, but boiling point H₂O > HF > NH₃.

Chapter 7 · Redox Reactions

Oxidation = loss of electrons = **increase** in oxidation number. **Reduction** = gain of electrons = **decrease** in oxidation number. **OIL RIG**.

The **oxidising agent is itself reduced**; the **reducing agent is itself oxidised**.

Oxidation number rules, in priority order

- Free element = **0** (Na, O₂, P₄, S₈, Cl₂)
- Monatomic ion = its charge (Na⁺ = +1, S^{2−} = −2)
- Fluorine is always −1** — no exceptions
- Group 1 = +1, group 2 = +2, Al = +3
- Hydrogen = +1, but −1 **in metal hydrides** (NaH, CaH₂, LiAlH₄)
- Oxygen = −2, except −1 **in peroxides** (H₂O₂), − $\frac{1}{2}$ **in superoxides** (KO₂), **positive with F** (+2 in OF₂)

7. Sum = 0 for a neutral molecule, = the charge for an ion

Worked values: Cr in $\text{Cr}_2\text{O}_7^{2-} = +6$ · Mn in $\text{MnO}_4^- = +7$ · **Fe in $\text{Fe}_3\text{O}_4 = +8/3$**
(average) · **S in $\text{S}_4\text{O}_6^{2-} = +2.5$** (average) · C in $\text{C}_6\text{H}_{12}\text{O}_6 = 0$.

A **fractional** oxidation number always means an **average** over inequivalent atoms.
Oxidation number may be positive, negative, zero or fractional; **valency is always a whole number with no sign.**

Types of redox reaction

Type	Example
Combination	$\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$ · $3\text{Mg} + \text{N}_2 \rightarrow \text{Mg}_3\text{N}_2$
Decomposition	$2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$ · $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$
Displacement (metal)	$\text{Zn(s)} + \text{CuSO}_4\text{(aq)} \rightarrow \text{ZnSO}_4\text{(aq)} + \text{Cu(s)}$
Displacement (non-metal)	$2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$ · $\text{Cl}_2 + 2\text{KBr} \rightarrow 2\text{KCl} + \text{Br}_2$
Disproportionation <i>same element both oxidised and reduced</i>	$2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$ (O: $-1 \rightarrow -2$ and 0) $\text{Cl}_2 + 2\text{OH}^- \rightarrow \text{Cl}^- + \text{ClO}^- + \text{H}_2\text{O}$ · $2\text{Cu}^+ \rightarrow \text{Cu}^{2+} + \text{Cu}$ $\text{P}_4 + 3\text{OH}^- + 3\text{H}_2\text{O} \rightarrow \text{PH}_3 + 3\text{H}_2\text{PO}_2^-$

DISPROPORTIONATION REQUIRES AN INTERMEDIATE OXIDATION STATE

Fluorine cannot disproportionate — it is the most electronegative element and has **no positive oxidation state**, so it can only be reduced from 0 to -1 .

Balancing — the two methods

METHOD 1 Oxidation number method

1. Assign oxidation numbers; identify which elements **change**.
2. Compute the increase and decrease **per atom**, then per formula unit.
3. Multiply so that **total increase = total decrease**.
4. Balance all other atoms except H and O by inspection.
5. Balance **O** with H_2O , then **H** with H^+ .
6. In basic medium, add OH^- to **both** sides equal to the H^+ , and combine into H_2O .
7. Check both **atoms and total charge**.

METHOD 2 Half-reaction (ion-electron) method

1. Split into the **oxidation half** and the **reduction half**.
2. Balance all atoms **except H and O**.
3. Balance **O** by adding H_2O .
4. Balance **H** by adding H^+ .
5. Balance **charge** by adding electrons.
6. Multiply so the **electrons cancel**, then add the halves.
7. **Basic medium**: add OH^- to both sides equal to the H^+ , convert $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$, cancel duplicated water.

Worked: $6\text{Fe}^{2+} + \text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ \rightarrow 6\text{Fe}^{3+} + 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$ (charge $+24 = +24$ ✓)

$2\text{MnO}_4^- + 5\text{SO}_3^{2-} + 6\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 5\text{SO}_4^{2-} + 3\text{H}_2\text{O}$ (charge $-6 = -6$ ✓)

$2\text{MnO}_4^- + 6\text{I}^- + 4\text{H}_2\text{O} \rightarrow 2\text{MnO}_2 + 3\text{I}_2 + 8\text{OH}^-$ (basic; charge $-8 = -8$ ✓)

Standard half-reactions and n-factors

Reagent / medium	Half-reaction	n
KMnO₄ — acidic (dil. H_2SO_4)	$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	5
KMnO ₄ — neutral / faintly alkaline	$\text{MnO}_4^- + 2\text{H}_2\text{O} + 3\text{e}^- \rightarrow \text{MnO}_2 + 4\text{OH}^-$	3
KMnO ₄ — strongly alkaline	$\text{MnO}_4^- + \text{e}^- \rightarrow \text{MnO}_4^{2-}$	1
K₂Cr₂O₇ — acidic only	$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	6
Iodometry (thiosulfate)	$\text{I}_2 + 2\text{S}_2\text{O}_3^{2-} \rightarrow 2\text{I}^- + \text{S}_4\text{O}_6^{2-}$	1 per $\text{S}_2\text{O}_3^{2-}$

ACIDIFY PERMANGANATE WITH DILUTE H_2SO_4 ONLY

HCl would itself be oxidised to Cl_2 , consuming extra KMnO_4 ; HNO_3 is itself an oxidising agent and would attack the reducing agent being estimated.

Titration arithmetic and electrode processes

EQUIVALENT MASS

$$E = \frac{\text{molar mass}}{n\text{-factor}}$$

NORMALITY †

$$N = M \times n\text{-factor}$$

AT THE END POINT †

$$N_1 V_1 = N_2 V_2$$

CELL EMF

$$E_{\text{cell}}^\circ = E_{\text{cathode}}^\circ - E_{\text{anode}}^\circ$$

positive $\Rightarrow$ spontaneous

Cell notation: $\text{Zn(s)} \mid \text{Zn}^{2+}(1 \text{ M}) \parallel \text{Cu}^{2+}(1 \text{ M}) \mid \text{Cu(s)}$ — **anode left, cathode right**, double line = salt bridge.

Daniell cell: $E^\circ = 0.34 - (-0.76) = +1.10 \text{ V}$. Anode is **negative** in a galvanic cell and is where **oxidation** occurs. Electrons flow anode $\rightarrow$ cathode through the wire; the salt bridge carries ions the other way to keep both solutions neutral.

SHE: $\text{Pt} \mid \text{H}_2(1 \text{ bar}) \mid \text{H}^+(1 \text{ M})$, defined as exactly **0.00 V**. More **negative** E° = stronger **reducing** agent; more **positive** E° = stronger **oxidising** agent (F_2 , $+2.87 \text{ V}$, is the strongest).

Activity series (most reactive first): Li K Ba Ca Na Mg Al Zn Fe Ni Sn Pb (H) Cu Hg Ag Au. Any metal displaces every metal **below** it from solution. Copper cannot liberate H_2 from dilute acid because its E° is positive.

E° values are never multiplied by stoichiometric coefficients.

CBSE Class 11 Chemistry, rationalised NCERT (2023–24 onward). Chapters 1–4 of Part I and Chapter 7 of Part II. Items marked [†] are standard exam formulae not stated in the NCERT text.